H

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATION
Higher 2

CANDIDATE NAME: **ANSWERS**

CLASS:

BIOLOGY **9744/03**

Paper 3 Long Structured and Free-response Questions

19 September 2022

Additional Materials: Answer Paper

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question on the separate Answer Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do no use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
1	**25**
2	**12**
3	**13**
Section B	**25**
Total	**75**

Section A

Answer **all** the questions in this section.

1 Human lactase is a transmembrane protein located in the brush border of the human small intestines. In humans, the lactase gene, *Lct*, is found on chromosome 2. Transcription and translation result in the formation of pre-pro-lactase (1927 amino acids), which is cleaved (cut) into pro-lactase (1023 amino acids) and then converted into its mature form, lactase.

Fig 1.1 shows the schematic of processing and localization of human lactase translational product.

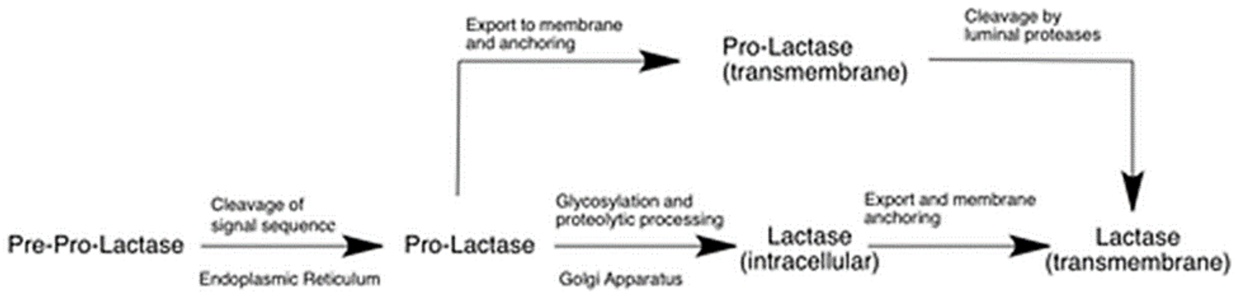


Fig. 1.1

(a) The primary translational product of the human *Lct* and rat lactase gene differ by only a single amino acid.
Suggest reasons for the similarity in the structure of human *Lct* and rat *Lct*.

any 2 of:

1 perform similar function/ bind to same substrate;

2 closely related / recent common ancestor;

3 lack of time for mutation to occur;

4 idea of, (if function very important) would be highly conserved;

[2]

(b) The pre-pro-lactase molecule is cleaved in the rough endoplasmic reticulum.
With reference to Fig. 1, describe the sequence of events occurring between cleavage in the rough endoplasmic reticulum and export of the molecule.

Ref to embedding of protein in vesicle membrane;

Ref to glycosylation and cleaving of amino acids (signal sequence);

packaged into vesicles by Golgi / budding off from Golgi;

vesicles move to / fuse with, cell surface membrane;

[3]

(c) Suggest why the signal sequence in pre-pro-lactase needs to be cleaved to form pro-lactase.

Signal sequence no longer required after protein is directed into rER;

Ref to signal sequence interfering with protein folding/ modification;

[2]

Research has shown that newborns express high levels of lactase. However, in most of the world's population, lactase transcription is down-regulated after weaning. Weaning is the process by which babies who were fully reliant on milk are introduced to solid foods.

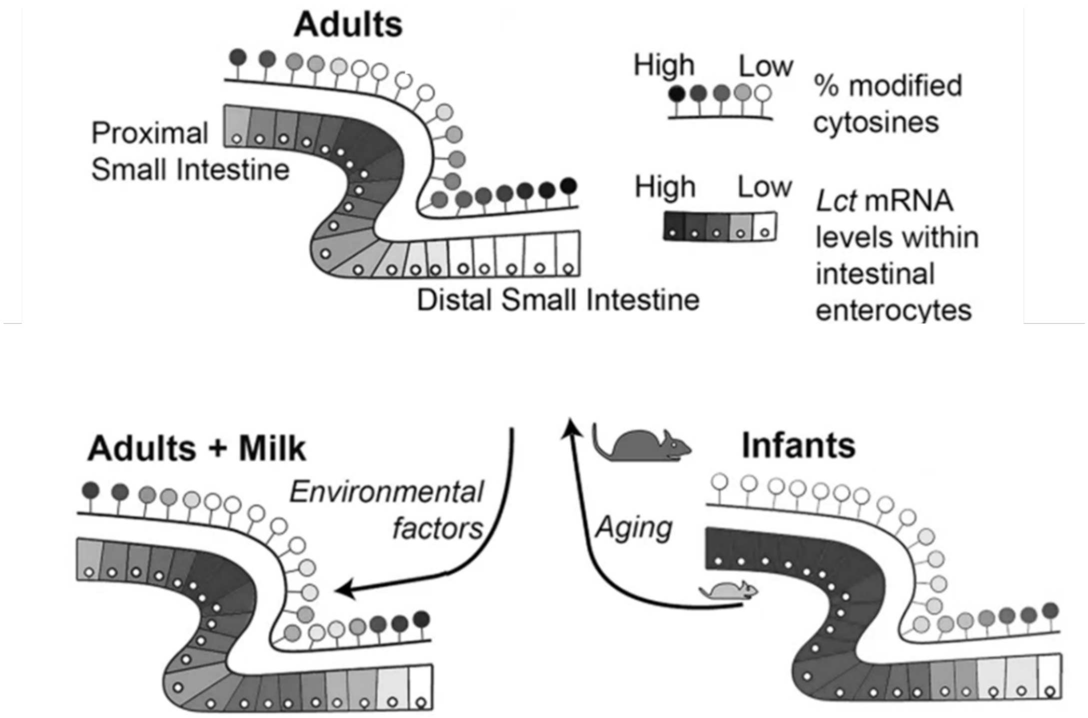


Fig. 1.2

(d) With reference to Fig. 1.2,

(i) Describe how the expression of *Lct* is influenced by environmental factors and by age.

Adults express overall lower levels of lactase in the intestines compared to infants;

Adults show a maximum of a moderate level of expression even in the proximal intestine vs infant shows a high level of expression in the proximal intestines;

Expression of lactase in adults is higher when milk is consumed as part of the diet compared to without;

[3]

(ii) Explain how the down-regulation of the gene expression occurred.

Percentage methylation of cytosines proportional to Lct mRNA levels/ higher percentage methylation of cytosines, lower mRNA levels;

(Methylation of cytosines) on DNA attracts histone deacetylases;

Deacetylated histones restores positive charges on lysine residues on histones;

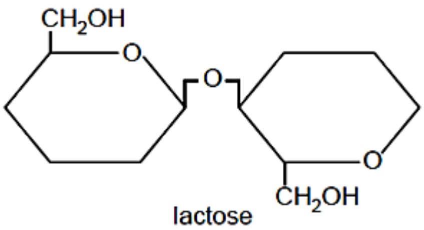
Increases condensation/ compaction/ interaction between negatively charged DNA and positively charged histones;

Reduces access by RNA polymerase, transcription decreases, less mRNA formed;

[3]

Lactose is the primary disaccharide in almost all mammalian milks. Cow and goat milk are some of the most common milk that is widely consumed all over the world. Some people become lactose intolerant because they cease to produce enough of the enzyme lactase. Lactase digests lactose to glucose and galactose.

(e) Complete Fig. 1.3 to show how lactose is broken down into glucose and galactose by the enzyme lactase.



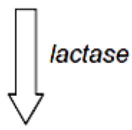


Fig. 1.3

1 glucose and galactose as separate molecules, at least one labelled ;
2 + water / H2O above the arrow ;
3 –H and –OH on position 1 of galactose and on position 4 of glucose ;

[3]

(f) Lactose-intolerant people suffer from severe discomfort, including diarrhoea, when they consume cow's milk or eat dairy products.

Fig. 1.4 shows the results of an investigation comparing:

- the effects on 50 lactose-intolerant volunteers of drinking normal cow's milk
- the effects on the same 50 lactose-intolerant volunteers of drinking low lactose cow's milk
- the effects on a control group of 15 volunteers, who were not lactose intolerant of drinking normal cow's milk.

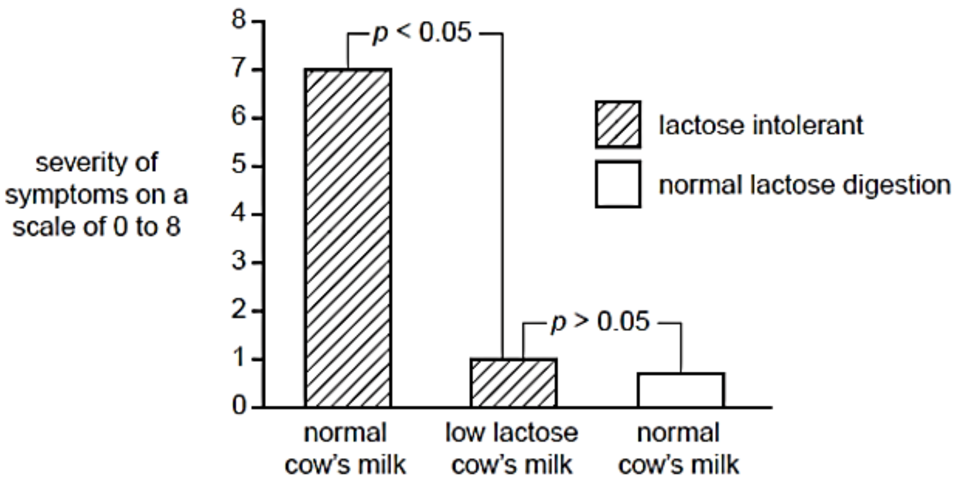


Fig. 1.4

With reference to the probability values shown in Fig. 1.4, comment on the results of this investigation.

$P < 0.05$:

(this difference is (statistically) significant)

low lactose milk causes much less severity of symptoms in lactose intolerant people than normal milk ;

$P > 0.05$:

(this difference is not (statistically) significant)

no difference in severity of symptoms between lactose-intolerant people drinking low lactose milk and non-lactose-intolerant people drinking normal milk ;

[2]

(g) Immobilised lactase can be used commercially to produce low lactose cow's milk from normal cow's milk for people who are lactose intolerant.

Discuss the advantages of using immobilised lactase rather than lactase that is not immobilised.

1. a small amount of enzyme can treat a large volume of milk ;
2. enzyme, remains active for longer / more stable ;
3. enzymes expensive to produce therefore immobilising enzymes, reduces costs / increases profits ;
4. enzyme can be, reused / recycled
5. downstream processing is easier / less contamination of product/ possible to have continuous process ;

[3]

(h) Due to an overlap with the symptoms, such as diarrhoea and vomiting, milk allergy is often confused with lactose intolerance. A food allergy happens when the immune system of a person overreacts to a specific food protein and it is associated with abnormally elevated antibody levels.

Describe how the presence of cow milk proteins induces the production of antibodies by B lymphocytes.

1. The B-cell receptor of a naive B lymphocyte binds to an intact antigen/ milk protein in the blood or lymph;
2. Antigen is taken up via endocytosis and digested into short peptides and presented on MHC;
3. T-cell receptor of the helper T cell binds to the peptide-MHC complex on the B lymphocytes;
4. (Resulting in) the secretion of cytokines by helper T cells for the activation of naïve B lymphocytes;
5. Upon binding to its antigen and interacting with helper T cell, the B lymphocyte divides and differentiates to produce plasma cells which produces and secretes antibodies;

[4]

[Total: 25]

2 Over thousands and millions of years, there have been natural cycles in the Earth's climate. There have been ice ages (e.g. the Quaternary Ice Age) and warmer interglacial periods. Climatic changes can be investigated using evidence left in tree rings, layers of ice in glaciers, ocean sediments and layers of sedimentary rocks. For example, bubbles of air in glacial ice trap tiny samples of Earth's atmosphere, giving scientists a history of greenhouse gases that stretches back more than 800,000 years.

These changes in climate affect plant distribution and physiology. Fig. 2.1 is a diagram showing the profile of two mountains in the tropics during a warm phase and a cool phase in the Earth's climate. The distribution of rain forest vegetation is also shown.

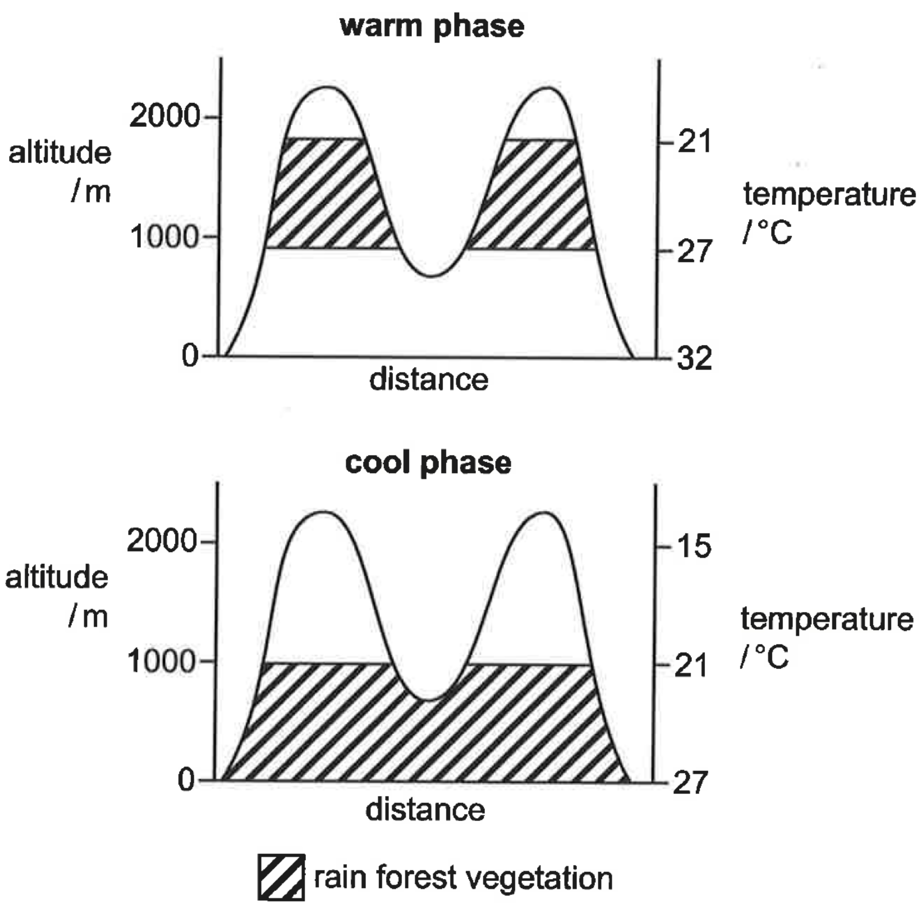


Fig. 2.1

(a)

(i) Describe and explain the effect of climate change on the distribution of rain forest vegetation in the tropics, as shown in Fig. 2.1.

Describe: 1. Warm phase 900—1800 m, cold phase 0—1000 m ; Lower temperatures at higher altitude/ORA ;

Describe: 2. Distribution continuous in cool phase, discontinuous/patchy/forming islands in warm phase ;

As temperature increases, the distribution of rain forest vegetation shifts towards higher altitudes/ORA ;

Rain forest vegetation will only grow in temperature range of 21—27 °C ;

[3]

(ii) Over millions of years there are repetitive cycles of alternating warm and cool phases in the Earth's climate.

Suggest how repeated changes in climate between the two phases shown in Fig. 2.1 may lead to greater species diversity.

Warm phase (max 2)

1. Geographical isolation leading to allopatric speciation ;
2. Reduced/no gene flow/interbreeding ;
3. Different conditions/different selection pressures ;

Cool phase

4. Ref. to spread of new species (to places that they were not prev found in) ;
5. previously isolated populations come back into contact but can no longer interbreed ; + accumulated genetic mutations
6. ref. to effect of repeated rounds of speciation/fragmentation ;

[4]

Changes in climate also affect insect physiology. Dengue virus is endemic in Singapore and the dengue virus vector is *Aedes aegypti*.

To study the influence of weather conditions on the incidence of dengue viral infection in Singapore, scientists recorded the amount of rainfall, mean weekly temperatures and the number of dengue cases over a one-year period. The recorded data is shown in Fig. 2.2.

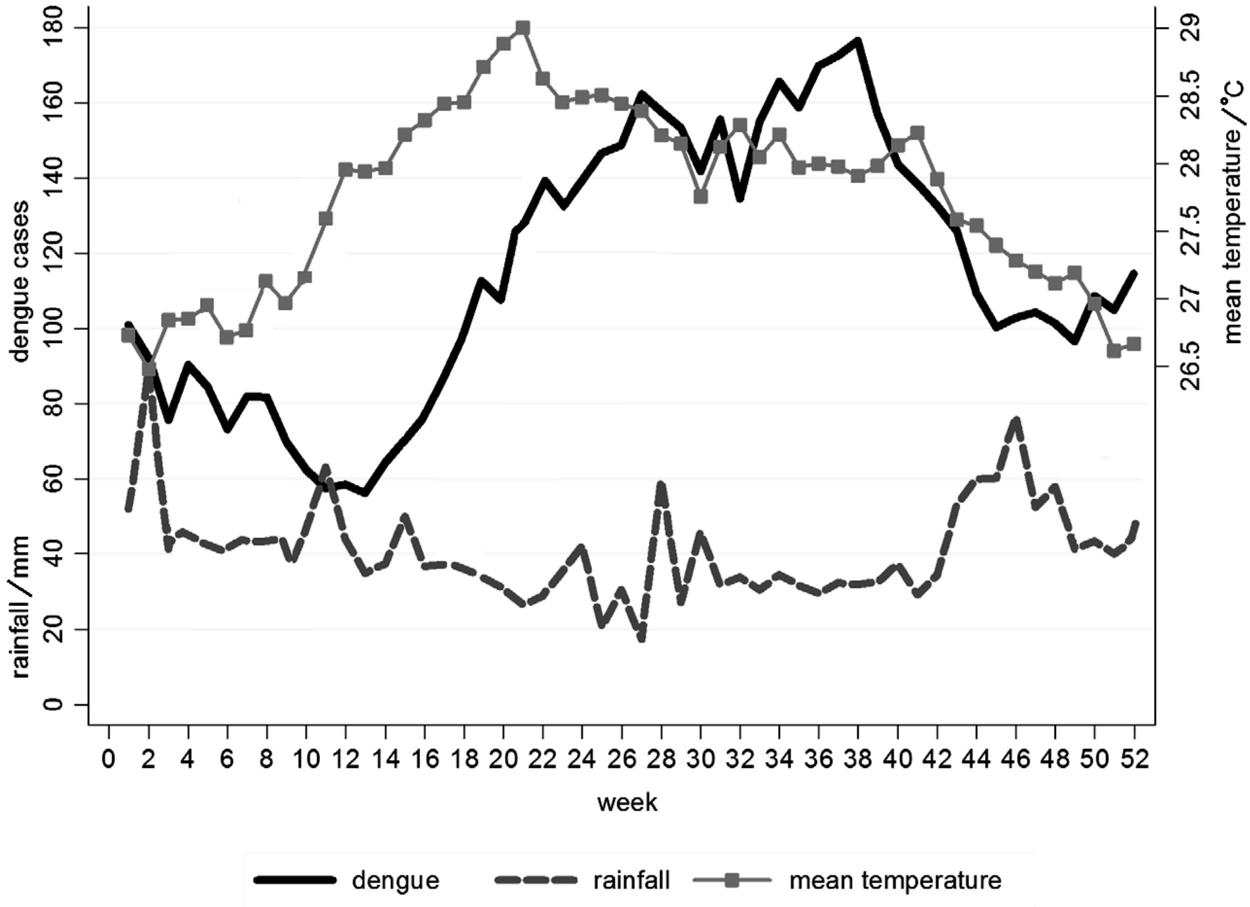


Fig. 2.2

(b) (i) Based on the information provided, comment on which weather condition is a better predictor of dengue cases in Singapore.

1. Mean weekly temperatures was a better predictor of the number of dengue cases;

2. Quoting relevant data to support correlation between temperature and dengue cases for upward trend e.g. When temperatures increased from 26.7oC at week 1 to 28.4oC at week 27, number of dengue cases increased from 100 to 160; OR

Quoting relevant data to support correlation between temperature and dengue cases for downward trend e.g.When temperatures decreased from 28oC at week 40 to 27.1oC at week 48, number of dengue cases decreased from 142 to 100;

3. Quoting relevant data to show that there is little correlation between rainfall and dengue cases (e.g. When rainfall remained relatively constant around 32mm

between week 32 and week 41, number of dengue cases increased sharply from 138 to 178, before decreasing to 140);

[3]

(b) **(ii)** Suggest explanations for the relationship between the weather condition chosen in part (i) and the number of dengue cases recorded.

1. Increase in temperature increases the rate of enzymatic activity and hence metabolism of Aedes mosquitos / mosquito vectors, increasing their activity and facilitating spread of dengue;

2. The development time of mosquito larvae shortens as temperature increases, increasing the number of mosquito vectors to spread the dengue virus;

3. Rate of dengue viral replication increases with increase in temperature, hence more infected individuals developing viremia for a longer period of time, facilitating spread of dengue;

R! Increase in precipitation brought about by increased temperature leads to the creation of standing water spots for the breeding of mosquitos (not supported by data)

[2]

[Total: 12]

3 Apoptosis is an important part of tissue homeostasis. It ensures that cells die quickly and are safely removed.

Every cell undergoing apoptosis shows a predictable series of changes. The cell shrinks and breaks up into dead cell fragments bounded by membranes. The membranes of these dead cell fragments display unique surface ligands, which identify the fragments to neighbouring cells ingesting these fragments.

Before the apoptotic cell breaks up, cellular DNA is degraded by endonucleases into small pieces that each differ in size by 180 base pairs.

Fig. 3.1 is an electron micrograph of a cell with ingested dead cell fragments.

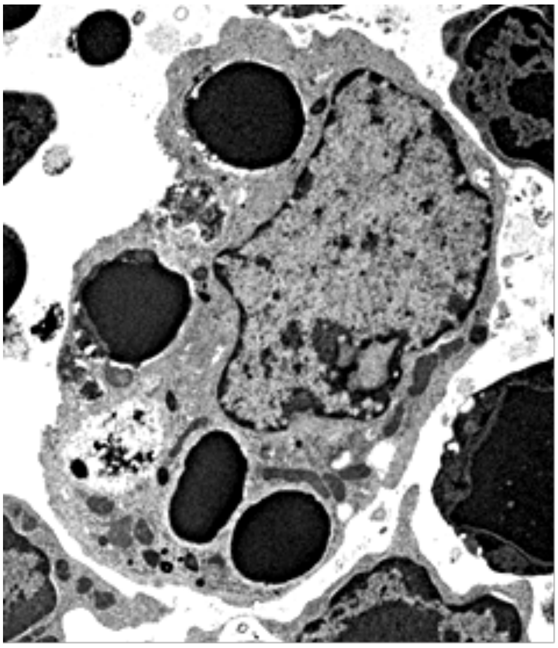


Fig. 3.1

(a)

(i) Name **one** type of cell that can ingest dead cell fragments and explain how the cell recognises these cell fragments.

- Name: macrophage, neutrophils;
- Idea that proteins on the outside of surface membrane of fragments expressed during apoptosis;
- These proteins are complementary in shape to the surface receptors on the macrophages.

[3]

(ii) Suggest how the DNA isolated from apoptotic cells can be analysed to confirm that apoptosis had indeed been activated.

GE + Separation based on size

DNA ladder is loaded into a separate well to add as a size reference;

Compulsory point

DNA bands (reject: fragments) that differs by 180bp would be observed

[3]

(b) Fig. 3.2 shows one way in which apoptosis can be triggered.

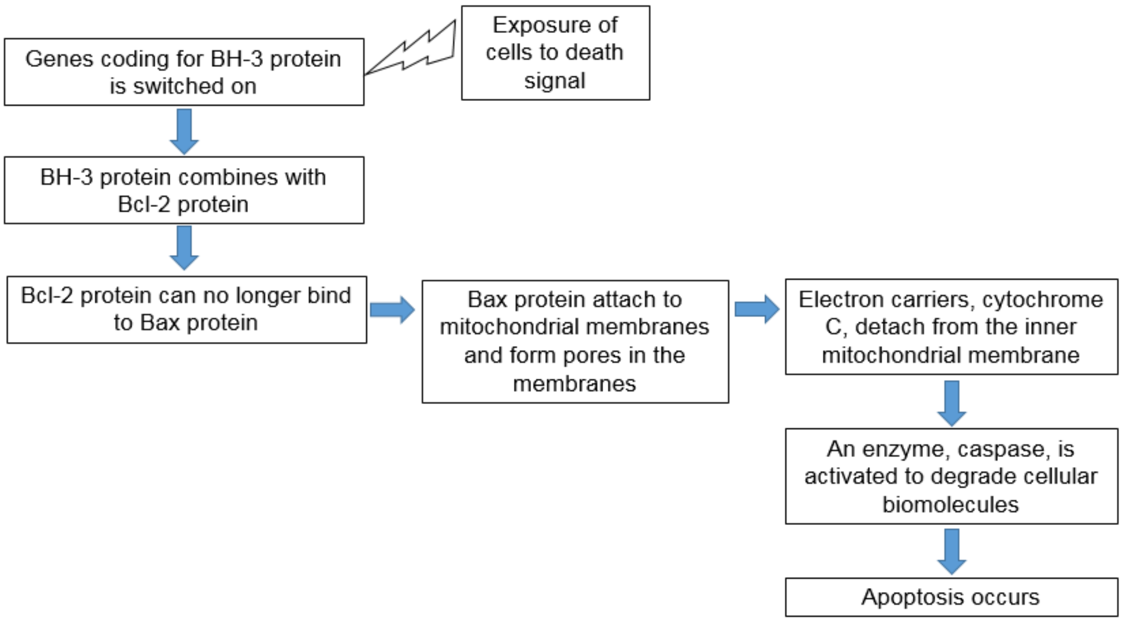


Fig. 3.2

(i) With reference to Fig 3.2, predict and explain how ATP production will change when cytochrome C is detached from the inner mitochondrial membrane.

- Lower production of ATP.
- Electrons flow through down the electron transport chain is disrupted.
- Proton gradient cannot be established across the inner mitochondrial membrane.

• Diffusion of H+ ions from inter-membrane space into matrix through ATP synthase cannot occur

[4]

(ii) A particular mutation leads to the excessive production of Bcl-2 protein, disrupting the apoptotic pathway, leading to B-cell lymphoma, a cancer of B-cells.

- Bcl-2 gene is found on chromosome 18.
- In B-cell lymphoma, a mutation shifts the Bcl-2 gene to chromosome 14, bringing it close to a powerful gene regulatory sequence.
- This regulatory sequence regulates the production of the immunoglobulin heavy chain.

Explain why the mutation will lead to B-cell lymphoma.

• Chromosomal translocation would place the bcl-2 gene under an active enhancer/promoter;

• Bcl-2 gene is transcribed and mRNA translated more often;

• High concentration of Bcl-2 protein would lead to decrease apoptosis/uncontrolled cell division;

[3]

[Total: 13]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

4 (a) Describe how genetic stability in eukaryotes is maintained at the molecular, cellular and population levels. [15]

(b) Discuss the importance of water in living organisms. [10]

[Total: 25]

5 (a) Cells are the smallest unit of life, and all their activities (processes and functions) are tightly controlled. [15]

Describe how cells control their activities and explain the importance of these controls.

(b) Discuss the importance of nitrogen in living organisms. [10]

[Total: 25]

4 (a)

Molecular level

- M1. Genetic stability at molecular level means maintaining same DNA sequence in daughter cells and parental cells;
- M2. In order to have identical copies of DNA as parent cell before nuclear division, DNA is first replicated during S phase of interphase via ***semi-conservative replication****;
- M3. Each parental strand of double stranded DNA act as ***template****;
- M4. To synthesise daughter strand following ***complementary**** base pairing;
- M5. Where nitrogenous base adenine forms 2 hydrogen bonds with thymine, guanine forms 3 hydrogen bonds with cytosine;
- M6. Each DNA molecule formed consists of one parental template with a daughter strand/newly synthesized strand;
- M7. ***DNA polymerase**** / part of it "proof-reads" newly synthesised region where if there is an incorrect DNA nucleotide is added, it will swiftly remove and replace with correct nucleotide. This is to ensure fidelity of DNA sequence;
- M8. Double stranded so that one of the strands can be a backup to be used as a template for repair;
- M9. Numerous hydrogen bonds of DNA increases stability of the molecule;
- M10. Telomeres at the ends of chromosomes prevent the loss of genetic information with each round of DNA replication;

Cellular level

- C1. Genetic stability at cellular level means maintaining same number and type of chromosomes in daughter and parental cells;

C2. During ***prophase****, identical DNA molecules condense to form genetically identical ***sister chromatids**** to allow for separation;

C3. During ***metaphase****, chromosomes arrange/align along ***metaphase plate****/at ***equator****;
R: middle of cell

C4. During anaphase, ***centromere**** divides and kinetochore microtubules from either pole attached to each sister chromatids, shortens;

C5. During ***anaphase****, genetically identical ***sister chromatids**** are separated and move to opposite ***poles**** ;
R: opposite ends

C6. Mitosis thus ensures equal distribution of identical sister chromatids to daughter cells;

C7. Cell cycle is regulated at ***checkpoints****. At these checkpoints, stop and go-ahead signals can determine to ensure the previous stage has occurred correctly before moving onto the next stage;

C8. The main checkpoints are at G_1, G_2 and M phase;

C9. When there is DNA damage, the *p53* gene is activated to produce p53 proteins to upregulate the expression of some genes;

C10. Cell cycle is halted so that there is enough time for cell to repair its damaged DNA;

C11. When DNA damage is beyond repair, apoptosis or cell death is initiated to stop the production of mutant daughter cells;

C12. Length of telomeres limit the number of cell divisions (Cells with telomeres at critical length do not go through any further cell divisions)

Population level

P1. Genetic stability at population level means maintaining same frequency of alleles in the population;

P2. Genetic stability occurs when there is no change in allele frequency, hence no major evolutionary forces such as natural selection, mutation, gene flow, genetic drift and sexual selection.

P3. More prevalent in asexually reproducing populations where there is no fusion of (genetically-different) gametes.

Events that do NOT contribute to genetic stability in a population

P4. natural selection selects for the same favourable alleles: individuals with hence allele frequency will NOT increase and contribute to genetic stability.

P5. Constant unchanging environment leads to no change in selection pressure.

P6. NO gene flow: there is NO transfer of alleles from one population to another through eg: migration

P7. NO ***genetic drift****: there is NO a change in allele frequency due to chance events on random individuals tends to reduce genetic stability in populations.

P8. Random mating: In sexually-reproducing populations, individuals DO NOT preferentially choose mates thus there is NO random mixing of gametes and genetic stability occurs. So random mating contributes to genetic stability, whereas non-random mating does not.

P9. NO mutation, new alleles are NOT generated and genetic stability occurs.

P10. ***Meiosis**** produces haploid cells (reductive division). This allows the diploid state to be restored upon fusion of gametes during fertilisation;

AVP: any mention of genetically identical;

QWC: at least one correct mark point from each of the three categories (molecular, cellular, population).

4 (b)

Hydrolysis:

- hydrolysis of any named bond peptide bonds / glycosidic / ester bonds involving one molecule of water per bond + enzyme
- Peptide bonds in polypeptides / Ester bonds in lipids
- Release glucose from glycogen / starch to be used as respiratory substrates by cells

Condensation

- Condensation + water released as by-product
- Importance: to form functional polymers + brief elab of specific function of named polymer / biomolecule ;;

Photosynthesis

- Photolysis of water in light dependent reactions
- Location of the reaction: chloroplast, plant cells
- Purpose: release electrons to fill up positive hole in PSII for non-cyclic photophosphorylation
- Ref to formation of ATP and red NADP for Calvin cycle to occur;
- Formation of sugars

Respiration

- Water as a by-product when oxygen acts as final electron and proton acceptor, catalysed by cytochrome oxidase
- Ref to ETC and ATP formation

Idea of aqueous medium for reactions to take place

- Ref to transport of substances, e.g. haemoglobin transporting O_2
- Ref to antibodies/ cytokines transported to site of infection/ circulating;

5 (a)

Control of cell division and importance (max 7)

- C1. Cell cycle **checkpoints** are regulated by **tumour suppressor gene and proto-oncogene products**;
- C2. Proto-oncogene products stimulate **normal cell growth and proliferation / cell division**;
- C3. Tumour suppressor gene products are involved in **inhibiting cell growth and division** / **activate cell cycle arrest**, DNA repair and/or apoptosis;
- C4. Cell cycle check points:
  - a. **G1 checkpoint** assesses if all organelles have been duplicated **before cell cycle is allowed to progress** into mitosis / meiosis;
  - b. **G2 checkpoint** assesses **if DNA replication is completed** and cell size is adequate, **before cell cycle is allowed to progress** into mitosis / meiosis;
  - c. **Metaphase checkpoint** assesses if **spindle fibres** are **attached** to **all chromosomes** before cell is allowed to enter **anaphase**;
- C5. **Loss-of-function mutation** in tumour suppressor genes and **gain-of-function mutation** in proto-oncogenes;
- C6. Ref to accumulation LOF and GOF mutations leading to **dysregulation of cell cycle checkpoints**;
- C7. **Dysregulation of cell cycle checkpoints** can result in **uncontrolled cell division**, leading to **tumour** formation;
- C8. Ref to abnormal cells continue to divide (because of dysregulation of checkpoints), allowing mutations to **accumulate**;

Control of gene expression in euk cells and importance (max 7)

- E1. In eukaryotes, **gene expression** is **upregulated** / turned "on" via **histone acetylation** / **chromatin remodelling complexes**;
- E2. **Chromatin** is **decondensed**, allowing **RNA polymerase** and **general transcription factors** to **bind** to **promoter** to form **transcriptional initiation complex**, allowing transcription to take place;
- E3. In eukaryotes, **gene expression** is **down-regulated** / repressed via **DNA methylation** / repressive **chromatin remodelling complexes**;
- E4. **Chromatin** is **condensed**, preventing **RNA polymerase** and **general transcription factors** to **bind** to **promoter**, and **transcriptional initiation complex** cannot form, preventing transcription;
- E5. This allows for **differential gene expression** when cells undergo **differentiation** / cells become specialised; (must have)
- E6. Ref to transcriptional control
- E7. Post transcriptional control + able to do alt splicing
- E8. Translational control
- E9. Post translational control
- E10. Idea of temporal control: producing proteins only when required
- E11. Idea of spatial control: producing proteins in different cell types;
- E12. Effective use of resources/ not wasting resources producing proteins not required;

Control of gene expression in prok cells (max 7)

- P1. In prokaryotes, all genes that code that enzymes of the same metabolic pathway are under the control of the same promoter in an operon.
- P2. Negative control is dependent on the state of the repressor. Operon is **switched "off"** when an active **repressor binds** to **operator**; (ORA)
- P3. Examples with elab such as Lac operon / Trp Operon;
- P4. Positive control is dependent on the state of the activator; Transcription is upregulated with an active **activator binds to the activator binding site;**
- P5. Example with elab: Catabolite Activator Protein activated by cAMP binds to CAP-binding site, stabilises transcription initiation complex;
- P6. Reference to translational repressors, controlling translation;
- P7. Production of polycistronic mRNA, whereby proteins required for the same metabolic pathway are encoded on the same mRNA, thus produced at the same time, improving efficiency;
- P8. Operons provide means for bacteria to produce proteins only when and where they are required, allowing the cell to conserve energy and resources;

Cell Signalling (max 5)

- CS1. ligand-receptor interaction, a specific ligand binds to the extracellular binding site of the receptor on the target cells and activates the receptor.
- CS2. During signal transduction, the activated receptor activates other relay molecules, usually a protein kinase. Activated receptor may also give rise to large amount of second messenger.
- CS3. Each activated protein kinase will initiate a sequential phosphorylation and activation of other kinases, resulting in a phosphorylation cascade.
- CS4. At each catalytic step in the cascade, the number of activated products is much greater than those in the preceding step resulting in signal amplification.
- CS5. The signal transduction pathway leads to a specific cellular response, which is the regulation of one or more cellular activities.
- CS6. Important in homeostasis, e.g. control of blood glucose level

Other points (Max 2)

- A1. Reference Activation of B cells/ T cells in adaptive immune response
- A2. [Membrane transport and export]: Polymerisation of microtubules to allow pseudopodia formation for movement/ movement of vesicle for secretion;
- A3. Ref to feedback inhibition e.g. PFK (rate determining step in glycolysis

5 (b)

Nitrogen as part of biomolecules:

- Amino acids to form proteins
- Nitrogenous base in nucleotides to form nucleic acids
- Choline in phospholipids

Proteins

- Ref to amino acid structure e.g. amino group
- Amino acid as monomers of proteins + primary structure

- Secondary structure
- Tertiary structure
- Quaternary structure
- Ref to named proteins, in structure vs function elab e.g. collagen → structural support
- Ref to globular enzymes → active sites @ antibody + specific antigen binding sites / Hb + binding sites / GPLR/RTK/ membrane-bound proteins

Nucleic acids

- Nucleotide structure (nitrogenous base)
- Monomer of DNA + brief elab of DNA as hereditary material ;;;
- Monomer of RNA + elab of role of mRNA, tRNA, rRNA ;;; (structure vs function points)
- Idea of CBP in replication, transcription, translation ;;
- Ref to universal genetic code based on nitrogenous bases

Role of mentioned biomolecules in living organisms:

Cell structure

- Ref to choline in phospholipids
- Ref to bilayer
- Functions of membranes
- Role of proteins in membranes

Respiration / Photosynthesis

- ATP / ADP structure + function
- Ref to co-enzymes NAD / FAD / NADP + function (max 1 for any of the named co-enzymes)
- Chlorophyll ring structure

AVP (Award marks if students' points not in syllabus but scientifically correct)

- Plant growth hormones